

CS & IT ENGINEERING



Algorithms

Greedy Method

Lecture No.- 07



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

MCST

Topics to be Covered



Topic

Topic

Topic

MCST

PYQs + Practice



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe

[NAT]

#Q. Consider the following undirected graph with edge weights as shown



22.22%

int(ans)

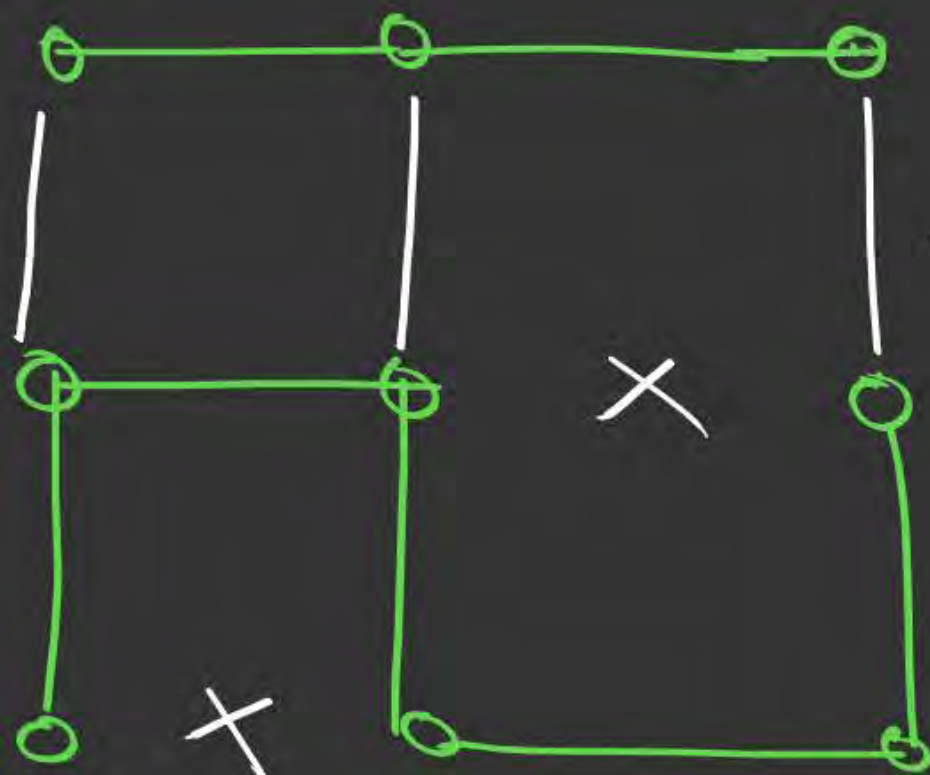
The number of minimum-weight spanning trees of the graph is _____.

Count

Ans = 3

$$n = 9$$

$$\begin{aligned} \text{edges in} \\ \text{MST} &= n - 1 \\ &= 8 \end{aligned}$$



$$\rightarrow {}^3C_1 = \underline{\underline{3}}$$

[MCQ]



Ticky → and $1 \leq y \leq n$

#Q. Consider a graph whose vertices are points in a plane with integer coordinates (x, y) where $1 \leq x \leq n$, $n > 2$ is an integer. 2 vertices $\langle x_1, y_1 \rangle$ & $\langle x_2, y_2 \rangle$ are adjacent iff $|x_1 - x_2| \leq 1$ and $|y_1 - y_2| \leq 1$. The cost of such an edge is given by the distance between them. Compute the weight of min cost spanning tree of such graph for a given value of n .

$n \times n$ grid

A n^2 X

B $(n^2 - 1) * 2$ X

C $n^2 - 1$

D $n(n - 1)$ $\Rightarrow 3 \times 2 = 6$ X

$n=3$ (8)

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Ans: C

63%

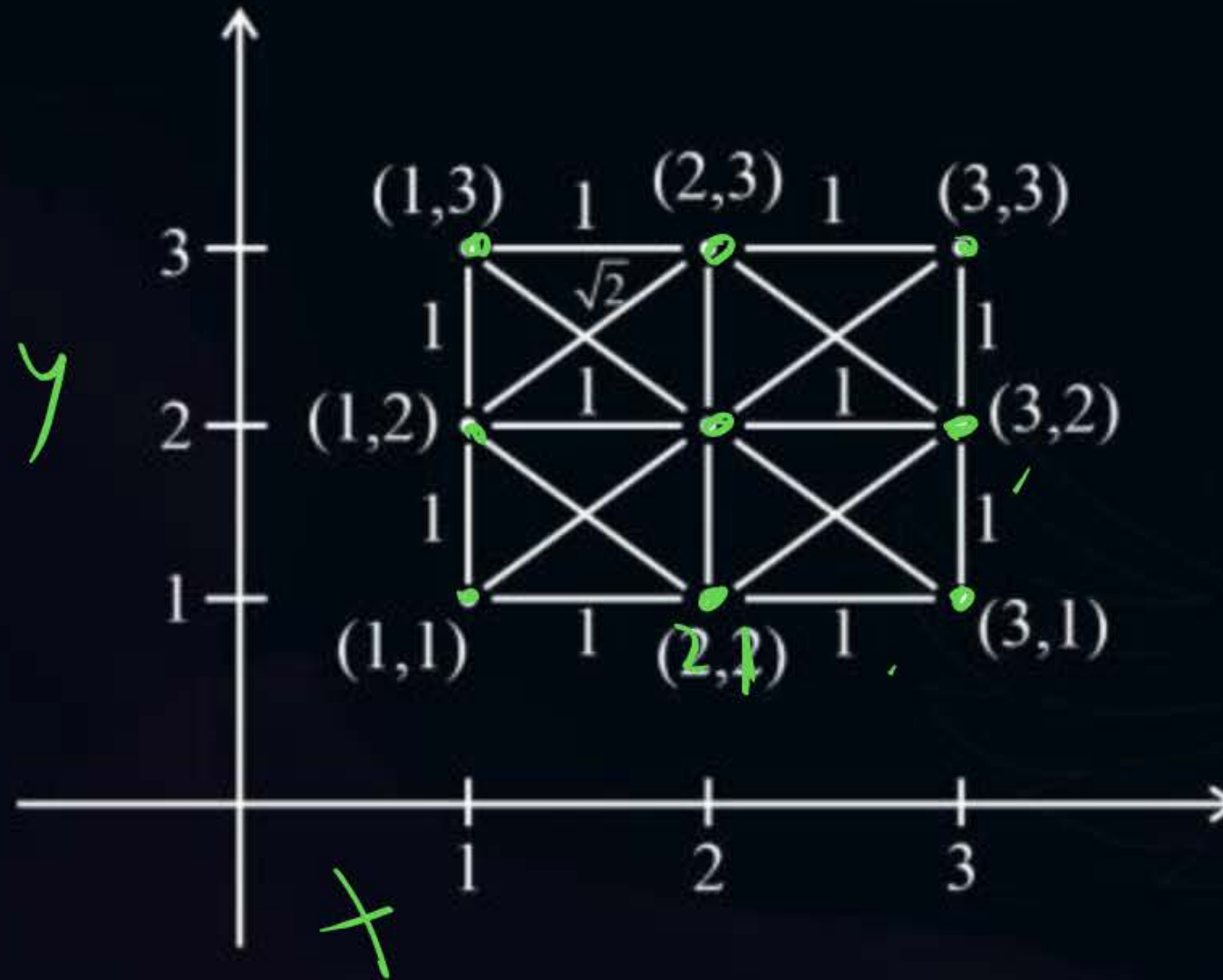
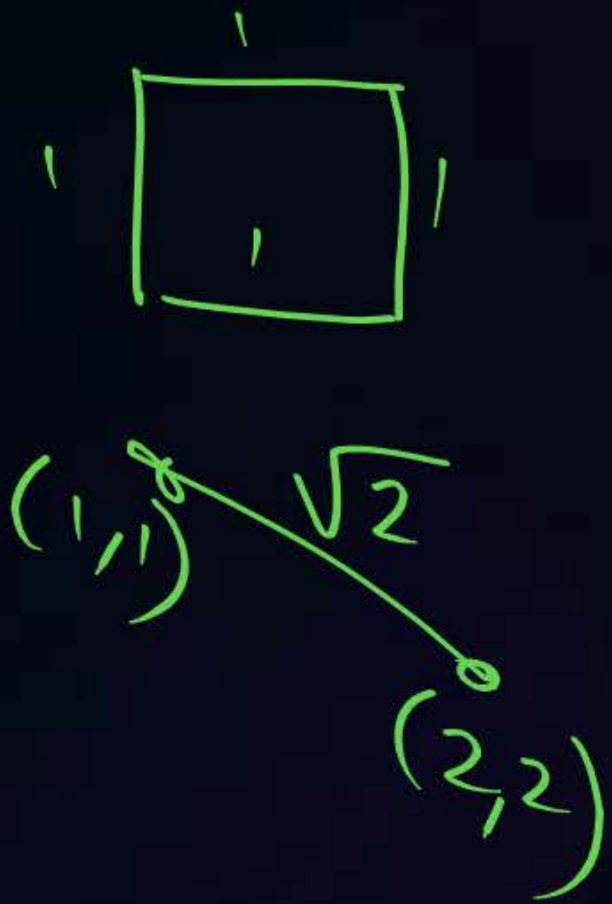


Topic : Greedy Algorithm



Solution:

$n = 3$



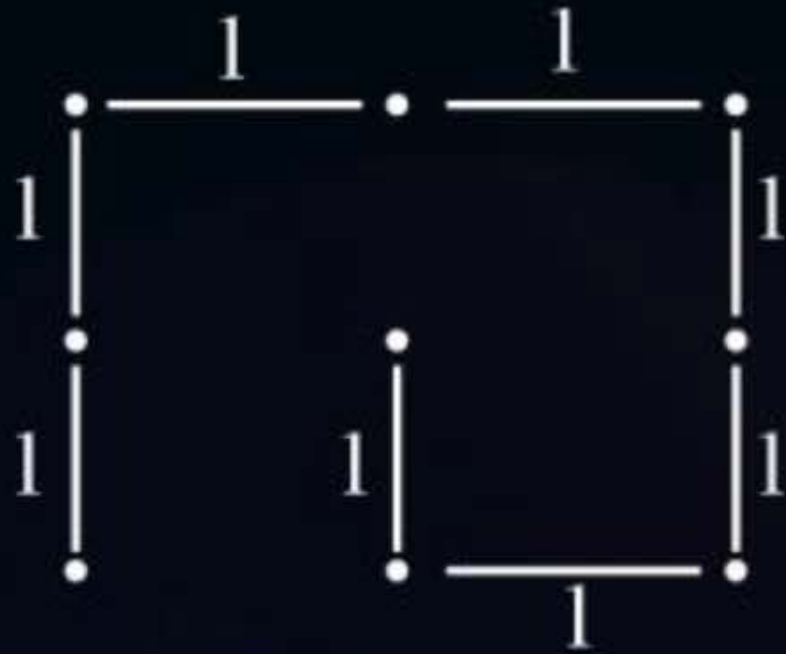
Cost of
mst = 8



Topic : Greedy Algorithm



Kruskal MCST



n

Vertices = n^2

edges in MCST = $(n^2 - 1)$

Cost of MCST = $(n^2 - 1) \times 1$
 $= (n^2 - 1)$

Friday: Cost of Max CST?

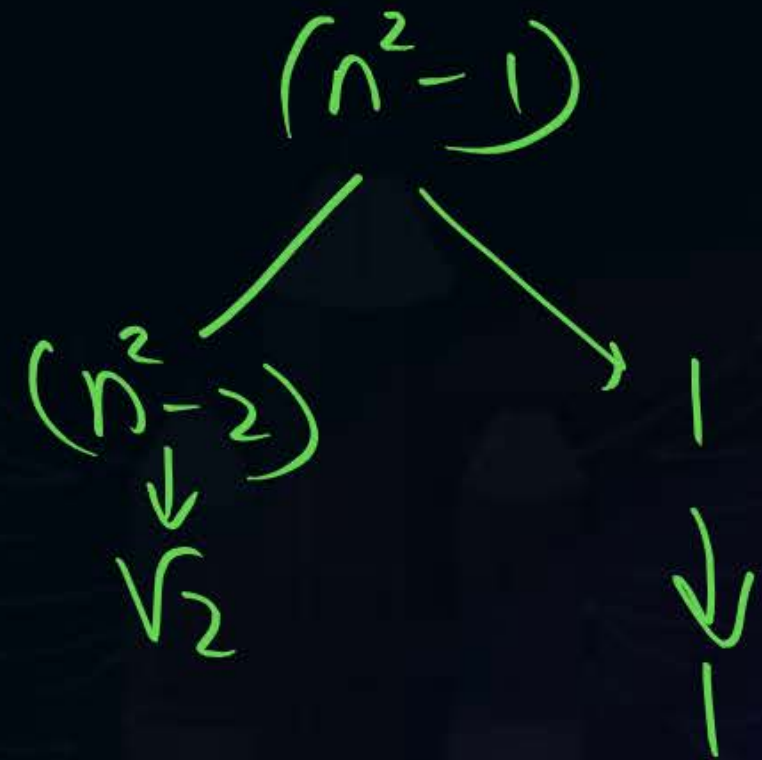
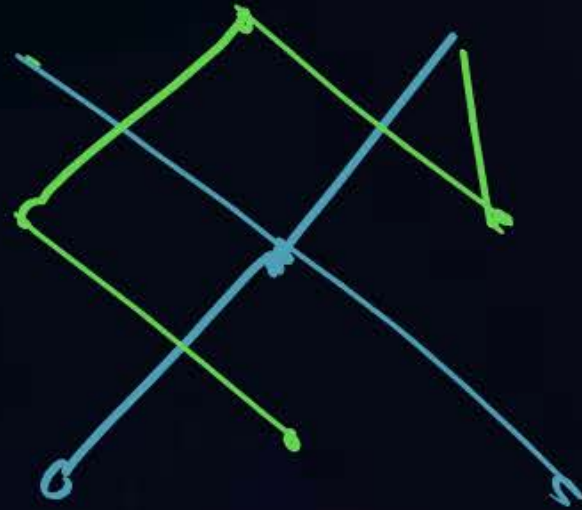


Topic : Greedy Algorithm



Variant 2:

#Q. What is the max cost possible for Spanning Tree?

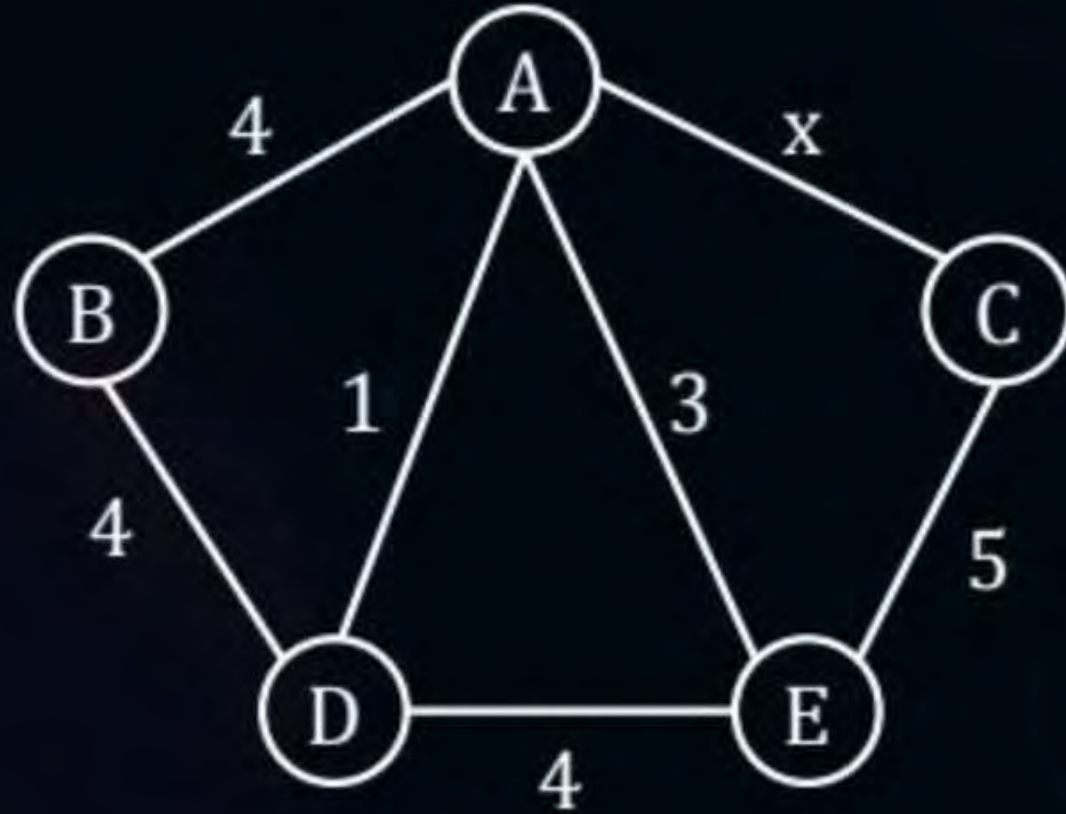


[NAT]

V.V. IITG



#Q. Consider the following undirected graph G.



75%

↓

38.5%

Count

Choose a value for x that will maximize the number of minimum weight spanning tree (MWSTs) of G . The number of MWSTs of G this value of x is _____.

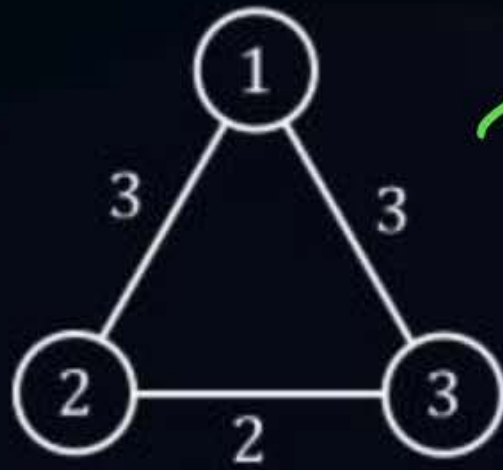
for



Topic : Greedy Techniques

Concept:-

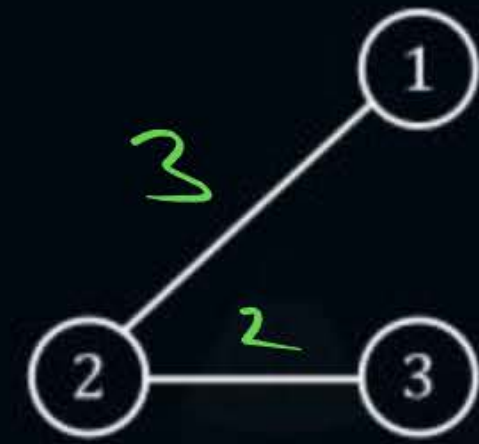
How many MCST?



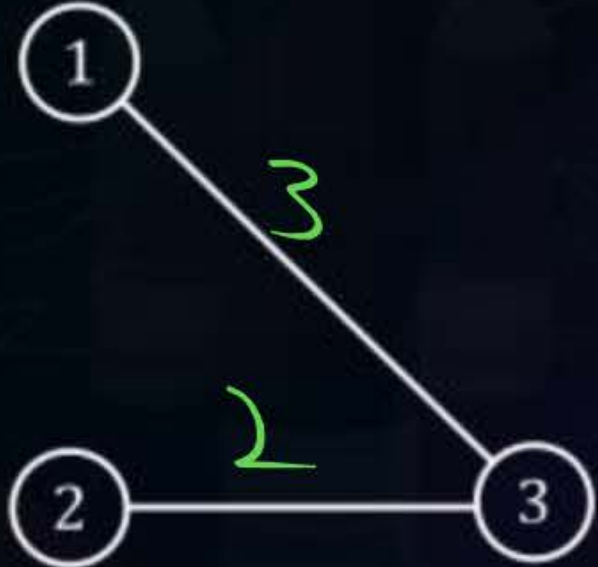
$${}^2C_1 = 2$$



${}^2C_1 = 2$



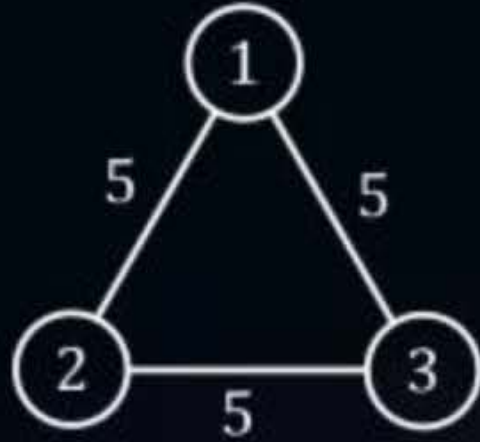
OR





Topic : Greedy Techniques

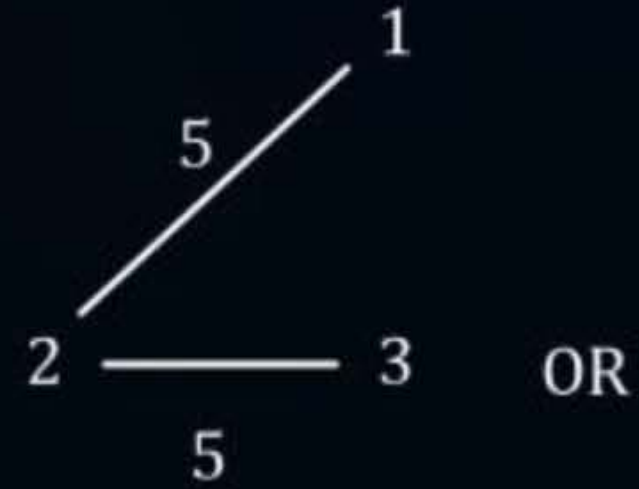
Eg.1.



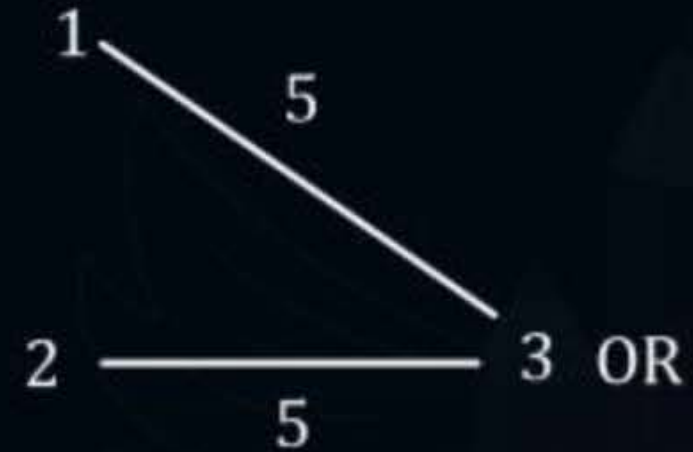
3C_2

3. MCSTs

$\rightarrow {}^3C_2 = \underline{\underline{3}}$



OR



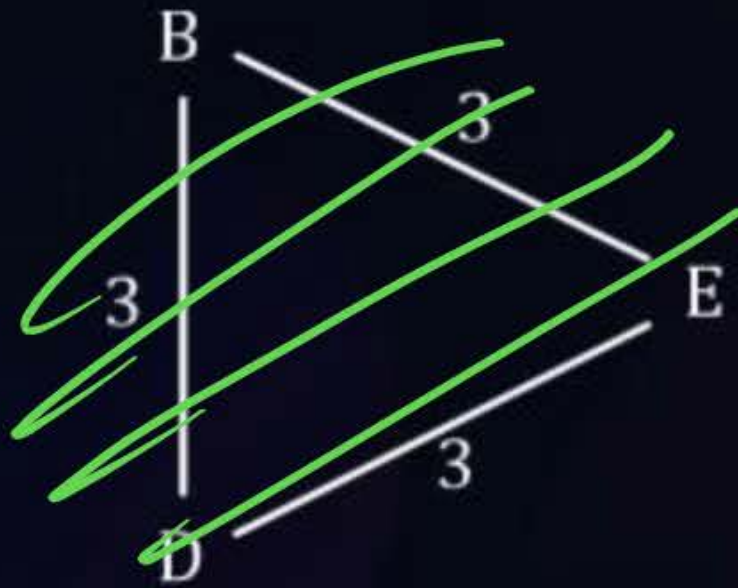
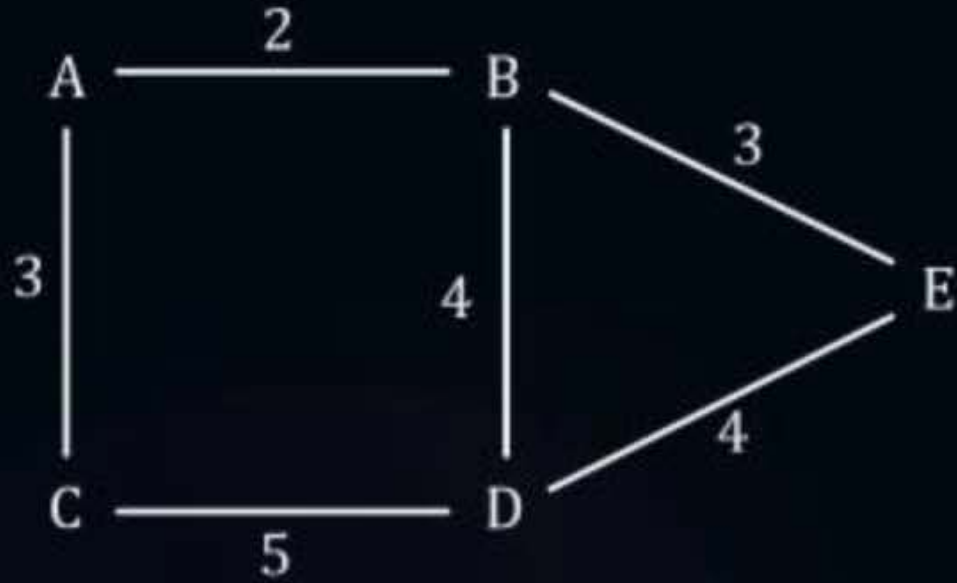
OR



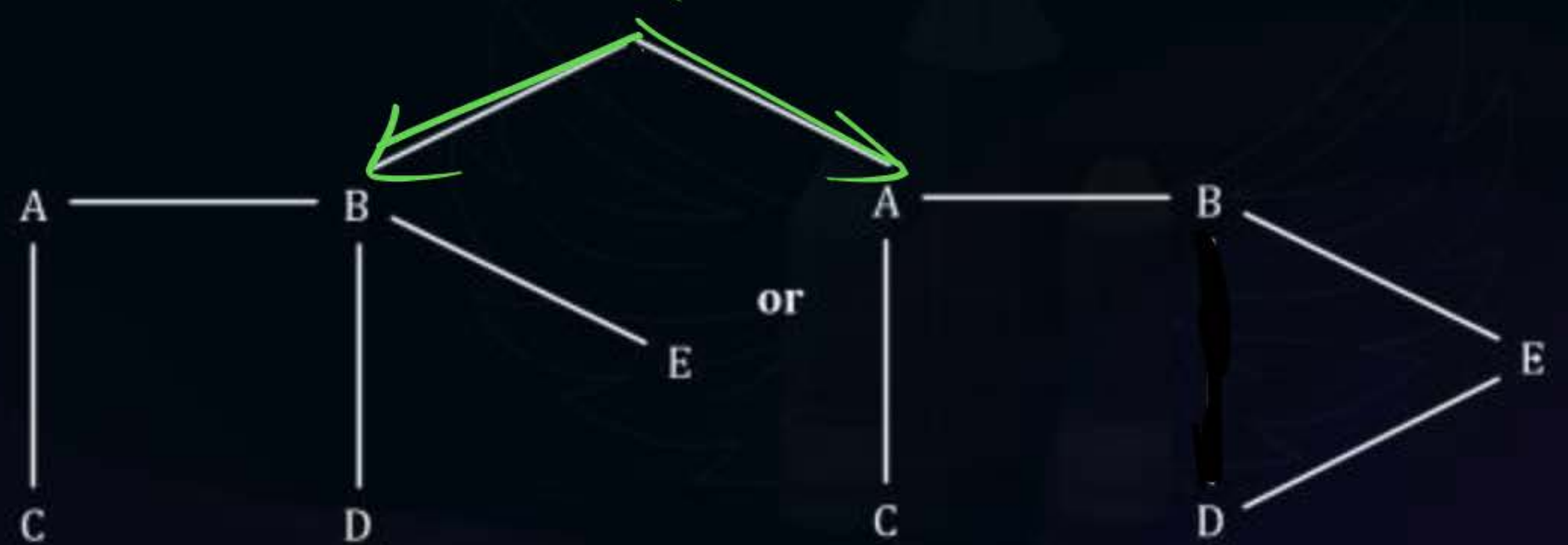
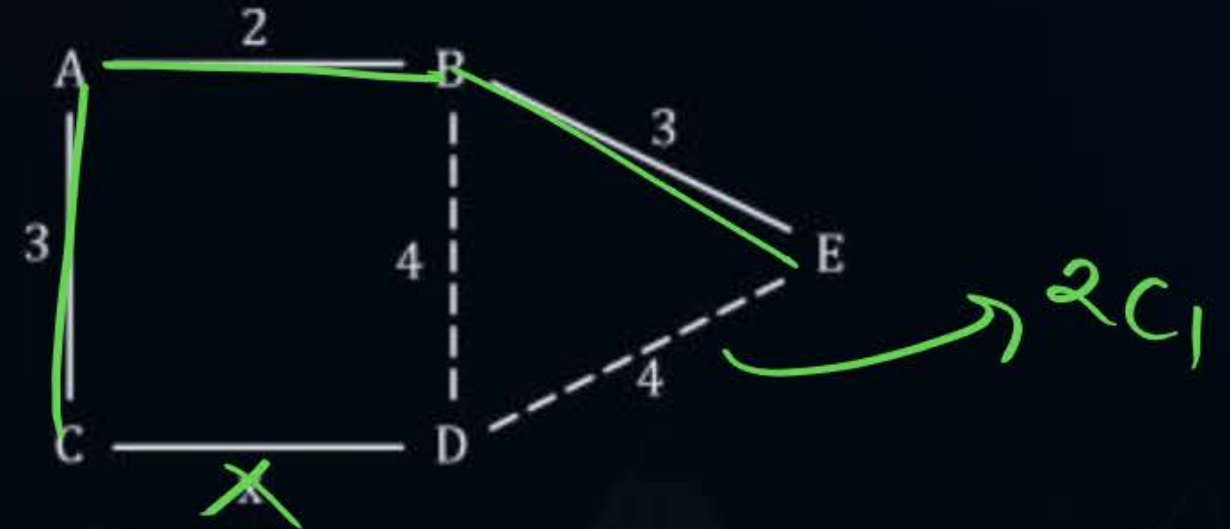


Topic : Greedy Techniques

Eg.2.



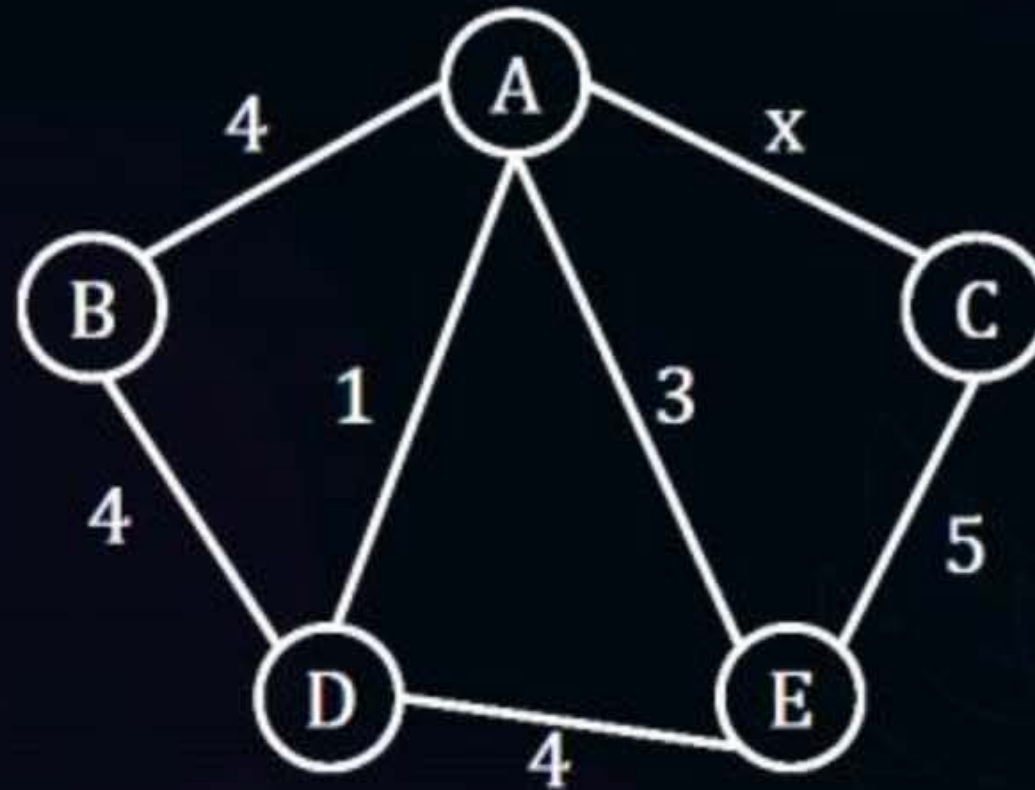
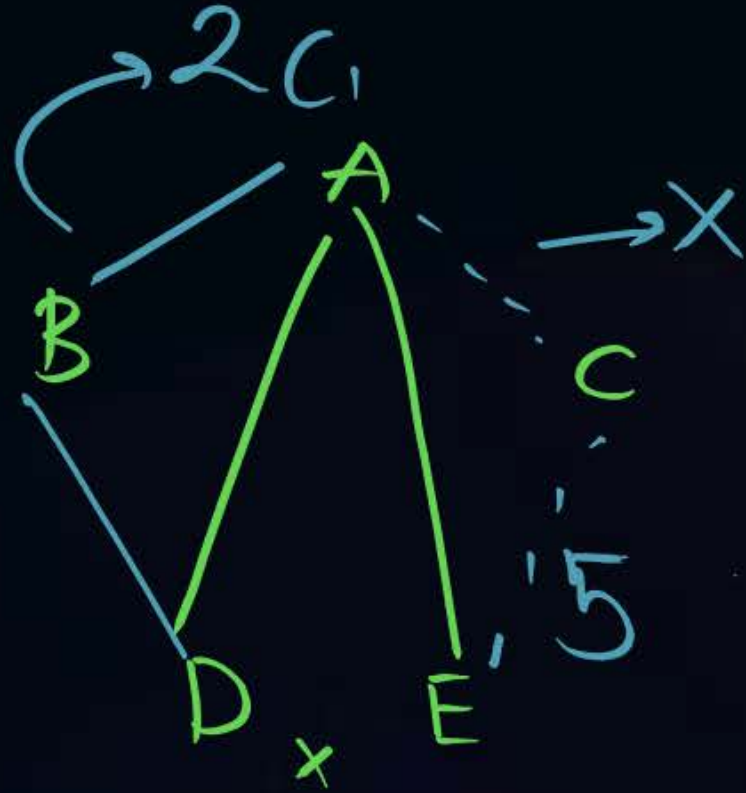
Kruskal





Topic : Greedy Techniques

#Q. Consider the following undirected graph G:



$$x = 5$$

$$\text{Count} = 4$$

Choose a value for x that maximize the number of minimum weight spanning tree (MWSTs) of G . The number of MWST of G this value of x is_____.

$$\text{Ans} = 4$$



Topic : Greedy Techniques

Possible Case:

1. $x > 5$: Then AC will never be a part ^{of} MCST and CE will be always a part.
→ Left → $2C_1$, Right → $1 \Rightarrow$ Total = ${}^2C_1 * 1 = \text{4 MCSTs}$ ~~2 MCSTs~~.

2. $x < 5$: Then AC will always be a ^{part} of MCSTs and CE will never be a part.
Left → 2C_1 , Right → $1 \Rightarrow$ Total ${}^2C_1 * 1 = \text{2 MCSTs}$

3. $x = 5$



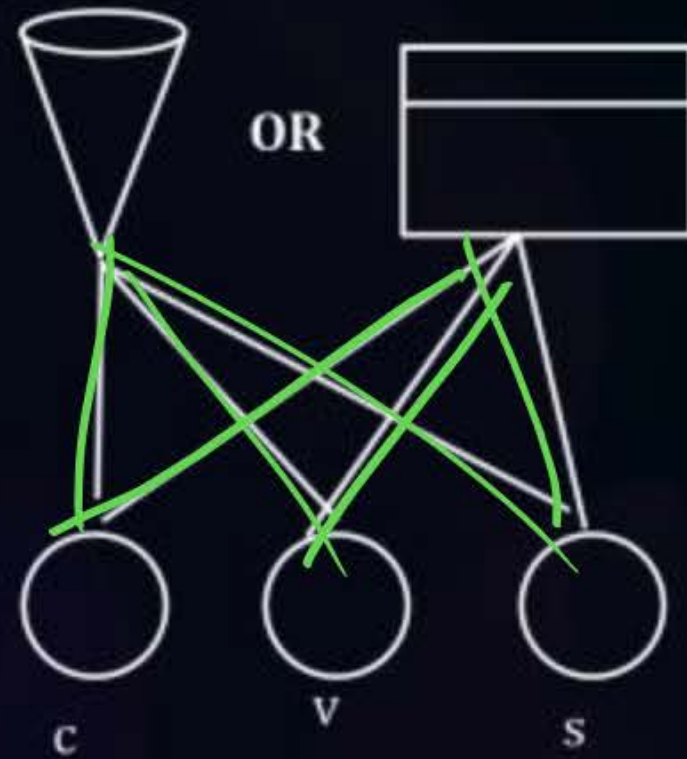
$\Rightarrow$ Right 2C_1 Left 2C_1 $\Rightarrow$ Total = ${}^2C_1 \times {}^2C_1$
 $= 2 \times 2$
 $= \text{4 MCSTs}$



Topic : Greedy Techniques

When to mail and add
Ice Cream

How many ways Ice cream can
be chosen?



$$2 \times 3 = 6$$



Topic : Greedy Techniques

P48

#Q. Let w be the minimum weight among all edge weights in an undirected connected graph. Let 'e' be a specific edge of weight ' w '. Which of the following is false?

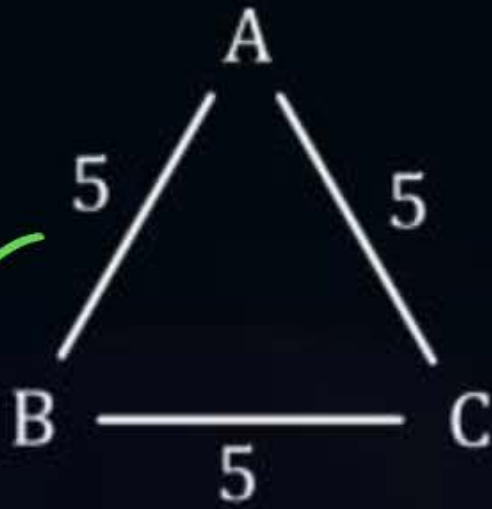
19.2%

- i. There is a minimum spanning tree containing 'e' always T
- ii. Every minimum spanning tree has an edges of weight ' w '. T
- iii. 'e' is present in every minimum spanning tree. F
- iv. If 'e' is not present in a minimum spanning tree named 'T' then there will be a cycle formed by adding 'e' to T. T



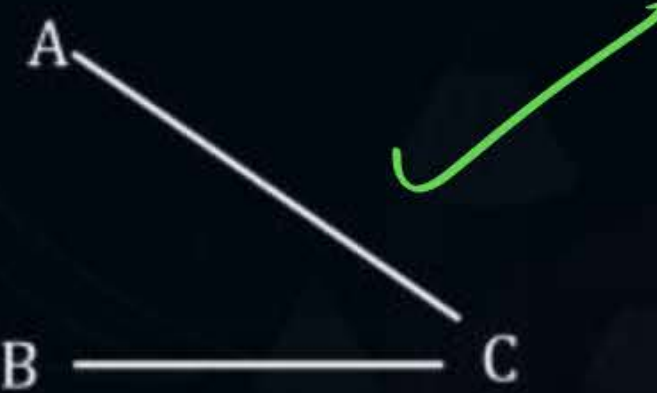
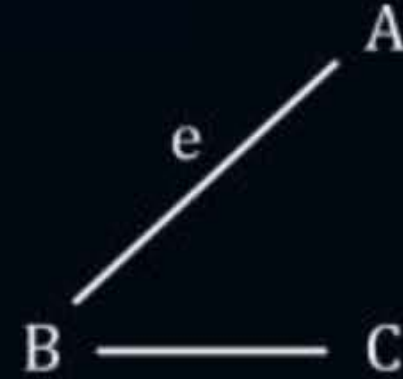
Topic : Greedy Techniques

Eg.3.

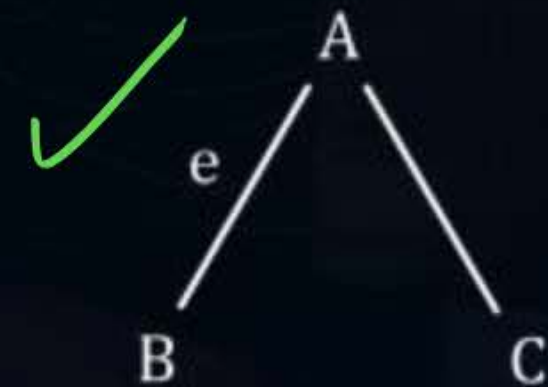
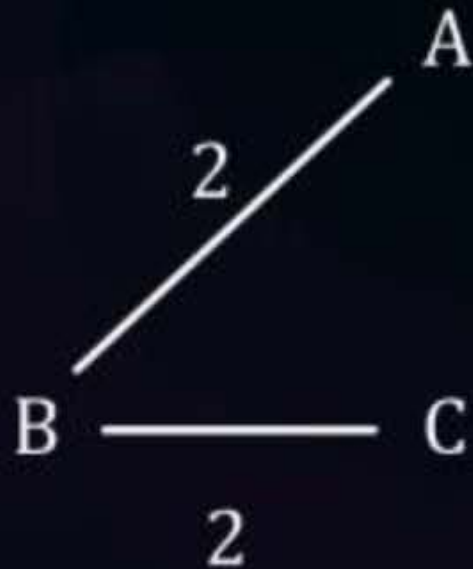
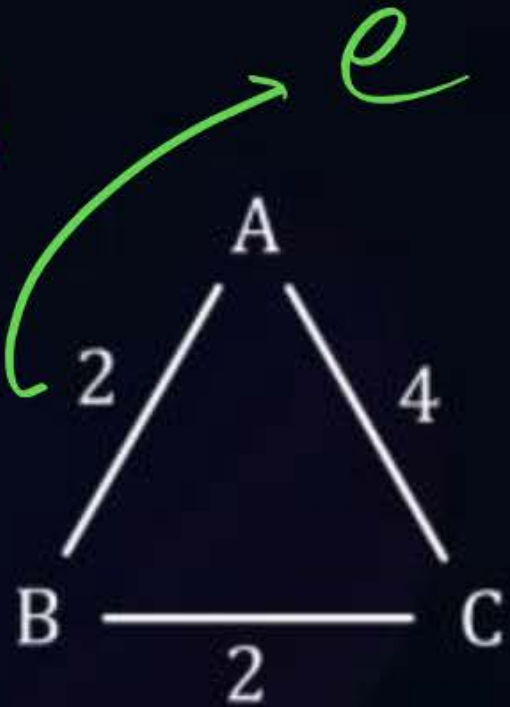


$W = 5$

$e = AB$



Eg.4.

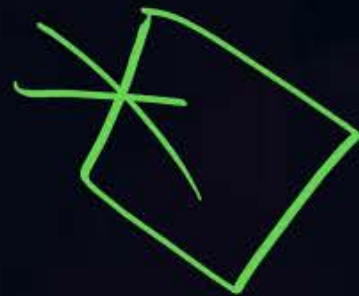




Topic : Greedy Techniques

#Q. $G = (V, E)$ is an undirected simple graph in ^{which} each edge has a distinct weight, and e is a particular edge of G . Which of the following statement about the minimum spanning tree (MSTs) of G is/are True? 26%

- I. If e is the lightest edge of some cycle in G , then every MST of G includes e .
- II. If e is the heaviest edge of some cycle in G , then every MST of G excludes e .



A

I only

C

Both I and II

B

II only

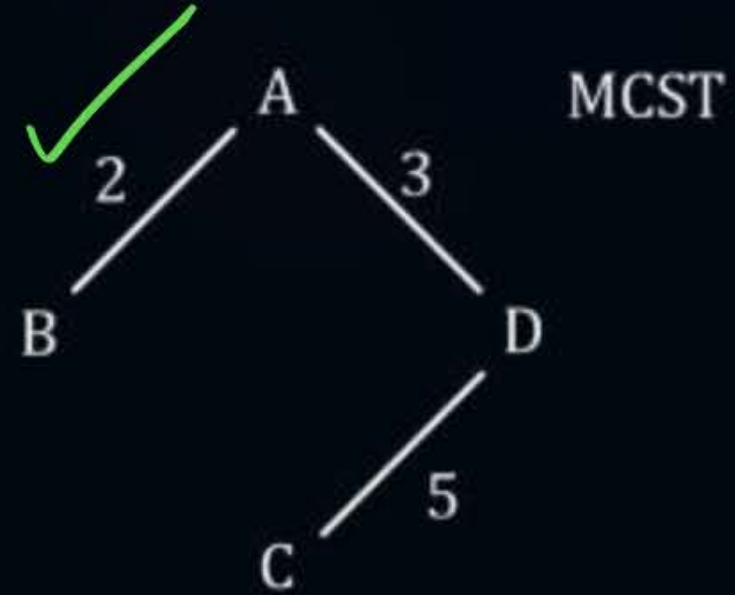
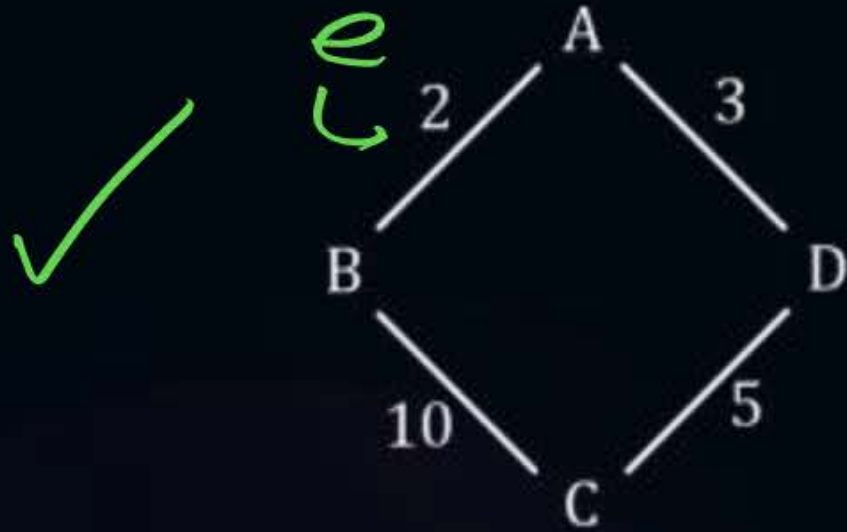
D

Neither I nor II

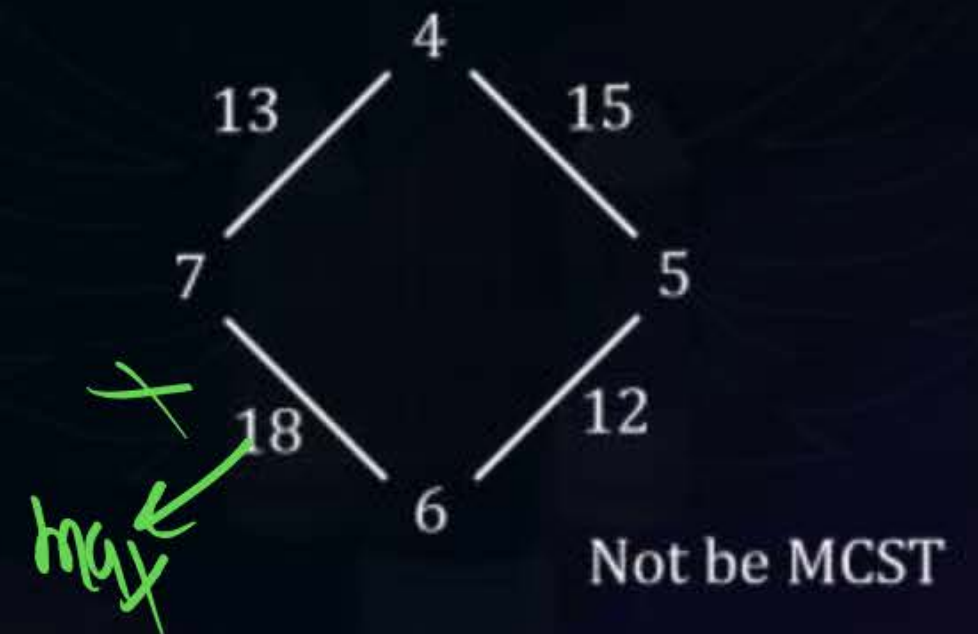
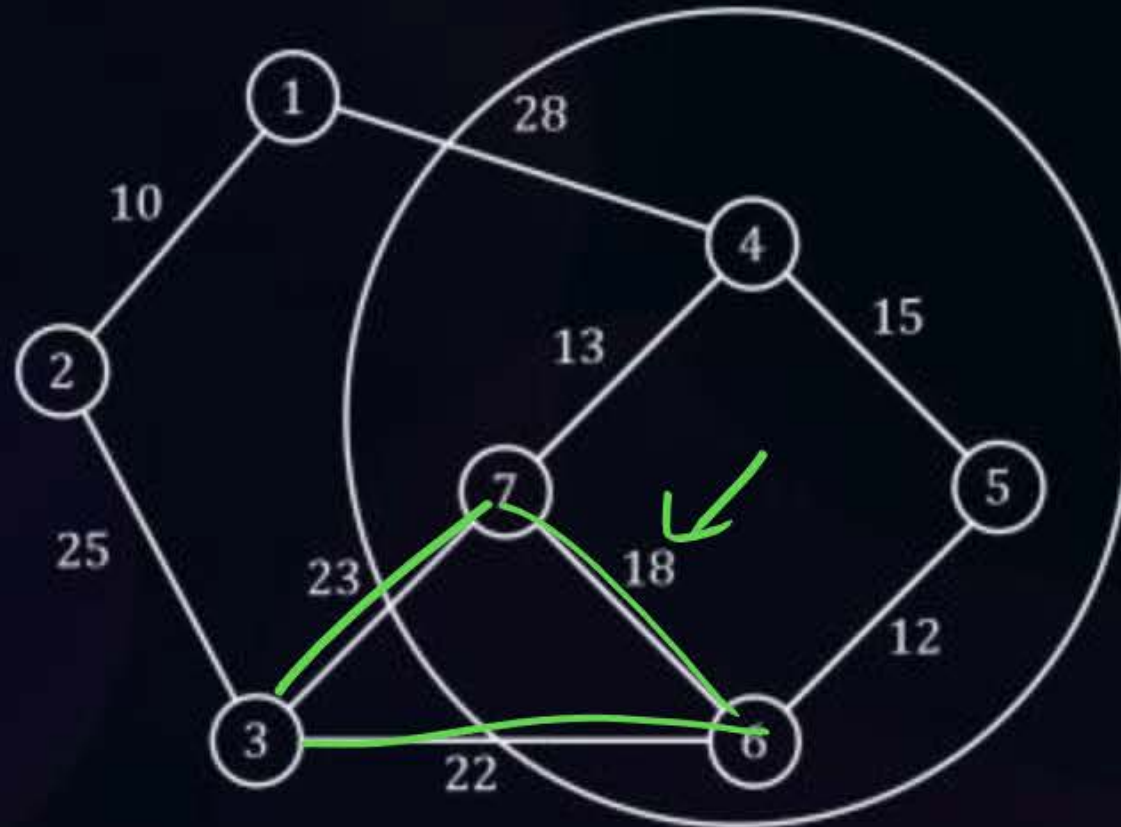


Topic : Greedy Techniques

Eg.7.



Eg.8.





Topic : Greedy Techniques

90%

two

63%

#Q. Let G be a connected undirect weight graph. Consider the following tow statements.

(F) S_1 : There exits a minimum weight edge in G which is present in every minimum spanning tree of G .

(T) S_2 : If every edge in G has distinct weight, then G has a unique minimum spanning tree.

A Both S_1 and S_2 are true

B S_1 is true and S_2 false

C S_1 is false and S_2 is true

D Both S_1 and S_2 are false



Topic : Greedy Techniques

HW

1. Algorithm Prime (E, cost, n, t)
2. {
3. Let (k,l) be an edge of minimum cost in E;
3. mincost: cost[k,l]
4. t[1,1]: =k; t[1,2]:1;
5. for i:=1 to n do // Initialize near.
6. If (cost [l, l]< cos [i,k]) then near [i] :=1;
7. else near [i]: =k;
8. near[k] := near [l] :=0;
9. For i:=2 to n-1 do



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HW

10. { // Find $n-2$ additional edges for t .
11. Let J be an index such that $\text{near}[j] \neq 0$ and
12. $\text{cost}[j, \text{near}[j]]$ is minimum;
13. $t[l,1] := j; t[i,2] := \text{near}[j];$
14. $\text{mincost} := \text{mincost} + \text{cost}[j, \text{near}[j]];$
15. $\text{near}[j] := 0;$
16. for $k := 1$ to n do // Update $\text{near}[]$.
17. If $((\text{near}[k] \neq 0) \text{ and } (\text{cost}[k, \text{near}[k]] > \text{cost}[k, j]))$
18. then $\text{near}[k] := j;$
19. }
20. Return $\text{mincost};$



Topic : Greedy Techniques

Time complexity Analysis of Prim's Algorithm;

1. Prve. Implementation (non-Heap based)

$$= e + n + c + m(n + c + n)$$

$$= O(n^2)$$

2. If heap is used in Prim's algo implementation

$$TC = (n + e) \log n$$

Ans: If $e \approx O(n^2)$ then $n^2 \log n$



Topic : Greedy Techniques



Kruskal code

Kruskal → End of course (Set + Heap)

Shortest Path Algos

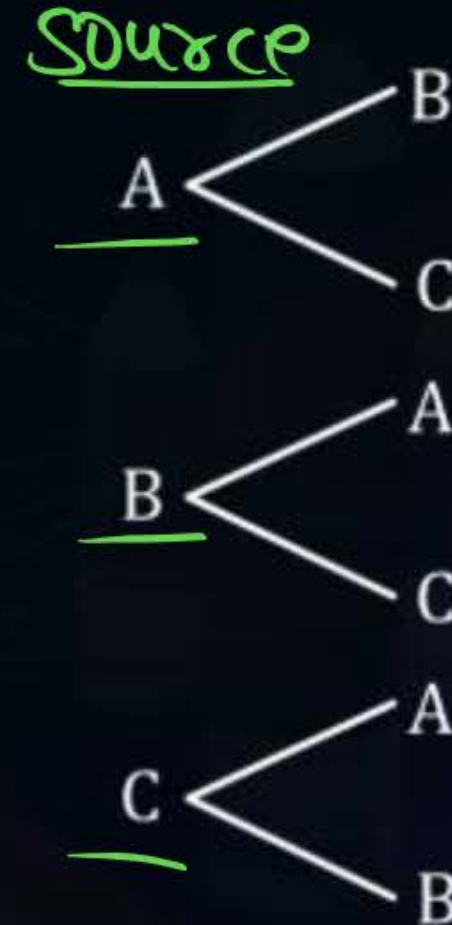
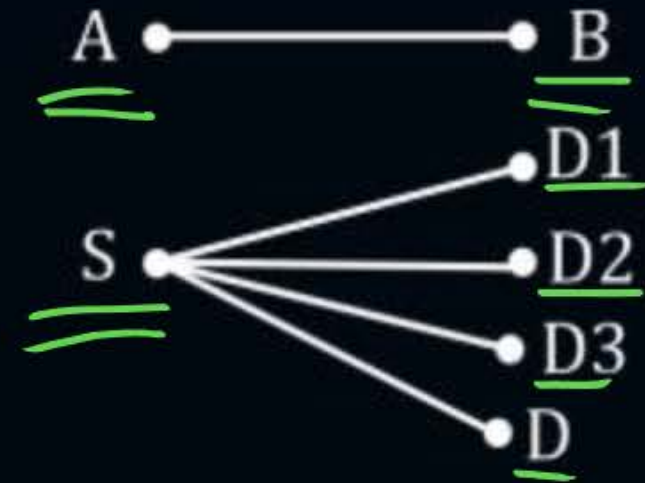
↓
Minimize Cost



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Shortest Path Algorithms Summary

1. Single Pair shortest Path (SPSP):
2. Single source shortest Paths (SSSP)
 - ★ (a) Dijkstra's SSSP Algo → Greedy
 - (b) Bellman ford → Dynamic Programming (DP)
3. All pairs shortest Path: (APSP)
 - (a) Floyd warshall algo
↓
Dynamic Programming (DP)





2 mins Summary



Topic

Topic

MCST

Topic

PYQs

Topic

Topic



THANK - YOU